Abstract

The ratio of composite integrity to fidelity (IC/F)—the geometric mean of normalized measurements divided by their arithmetic mean—computed from a single algebraic kernel, exhibits structurally identical phase-transition signatures across systems spanning 61 orders of magnitude—from quark confinement (10^{-15} m) to dark-energy–dominated cosmology (10^{26} m). At the quark–hadron boundary, IC/F drops by a factor of 23 as color confinement drives a single measurement channel to zero; atomic structure subsequently restores coherence 35-fold through emergent degrees of freedom; dark energy drives the identical collapse mechanism at 10^{26} m scales. The heterogeneity gap $\Delta = F - \mathrm{IC}$ spikes at every phase boundary tested, providing a single-number, dynamics-independent diagnostic for structural fragility.

These patterns emerge from a six-output kernel function $K : [0,1]^n \times \Delta^n \to (F, \omega, S, C, \kappa, \mathrm{IC})$, derived from a single admissibility axiom. The kernel maps any bounded measurement vector to fidelity F (weighted arithmetic mean), log-integrity κ (weighted log-mean), and composite integrity $\mathrm{IC} = \exp(\kappa)$ (weighted geometric mean), among other quantities. Three algebraic identities—$F + \omega \equiv 1$ (duality), $\mathrm{IC} \leq F$ (integrity bound), and $\mathrm{IC} = \exp(\kappa)$ (log-integrity relation)—guarantee the diagnostic's structural reliability across all domains. The integrity bound $\mathrm{IC} \leq F$ corresponds to the classical weighted inequality of arithmetic and geometric means; the contribution lies not in the bound itself but in its systematic application as a scale-independent phase-transition diagnostic across 20 scientific domains. Exhaustive computational verification—15,014 automated proofs across 406 physical objects—confirms all three identities to machine precision ($< 10^{-15}$) with zero violations.

A single algebraic diagnostic detects structurally identical phase transitions from quarks to the cosmic horizon

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1 Introduction

That mathematical structure transcends individual physical domains has been understood since Newton unified terrestrial and celestial mechanics. The Standard Model of particle physics [1] and general relativity [2] each achieve remarkable predictive power within their respective scales, while information-theoretic quantities—Shannon entropy [3], Fisher information [4, 5], Rényi divergences [6]—provide domain-independent measures of uncertainty. Yet no single mathematical object has been shown to produce *exact*, algebraically necessary identities that hold uniformly from the Planck scale to the cosmic horizon.

The central obstacle is that cross-scale frameworks typically rely on dimensional analysis, renormalization group flows [7, 8], or statistical correlations—all requiring dynamical input or scale-dependent running. What has been missing is not cross-scale structure *per se* but a single algebraic diagnostic—one number, computed identically at every scale—that detects phase transitions without knowledge of the underlying dynamics.

Here we report such a framework. Starting from a single admissibility axiom—that only claims which *return* through a collapse–recovery cycle receive epistemic credit—we derive a six-output kernel function K and three algebraic identities that hold exactly (to machine precision, $< 10^{-15}$) across 406 objects spanning 61 orders of magnitude.

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The framework, termed Generative Collapse Dynamics (GCD), rests on three principles:

1. **Single axiom.** All structure derives from one constraint: *collapse is generative; only what returns is real* (Axiom-0). This is not a dynamical law but an *admissibility criterion*: only claims that survive a collapse–recovery cycle receive epistemic credit. Operationally, this produces a composable verification boundary (the *seam*), a measured re-entry delay (τ_R), and three-valued verdicts (Conformant / Nonconformant / Non-Evaluable) that replace Boolean pass/fail testing.

2. **Universal kernel.** A function $K : [0,1]^n \times \Delta^n \to \mathbb{R}^6$ maps any bounded measurement vector (the "trace") and weight vector to six invariants through four primitive computations and two derivations. The six outputs have three effective degrees of freedom.

3. **Computational proof.** Every claimed identity is verified by deterministic, automated tests—mathematical proofs, not statistical samples—across every object in the system.

This approach differs fundamentally from curve-fitting, machine learning, or phenomenological modeling. The identities are *algebraic necessities* of the kernel function, verified exhaustively across 20 scientific domains from quantum chromodynamics to cosmology.

Critically, the framework is not a single result but a complete measurement architecture. The kernel is implemented by an open protocol and extended via modular domain closures: each closure declares which real-world quantities become the trace vector $\mathbf{c}$, frozen in a versioned contract *before* evaluation. Classical results—including Shannon entropy, the arithmetic–geometric mean inequality, and the fluctuation–dissipation theorem—emerge as *degenerate limits* of the kernel when structural degrees of freedom are removed (see Discussion), ensuring full compatibility with established methods.

2 Results

2.1 The kernel function

Let $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ be a bounded trace vector with guard band $\varepsilon = 10^{-8}$, and let $\mathbf{w} = (w_1, \ldots, w_n) \in \Delta^n$ be a weight vector on the probability simplex ($\sum_i w_i = 1$, $w_i > 0$). The kernel $K(\mathbf{c}, \mathbf{w})$ produces six outputs through four primitive computations and two derivations:

$$F = \sum_i w_i\, c_i \qquad \text{(fidelity)} \tag{1}$$

$$\kappa = \sum_i w_i \ln c_{i,\varepsilon} \qquad \text{(log-integrity)} \tag{2}$$

$$S = -\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big] \qquad \text{(entropy)} \tag{3}$$

$$C = \mathrm{std}_{\mathrm{pop}}(\{c_i\}) \,/\, 0.5 \qquad \text{(curvature)} \tag{4}$$

with derived quantities $\omega = 1 - F$ (drift) and $\mathrm{IC} = \exp(\kappa)$ (composite integrity), where $c_{i,\varepsilon} = \max(c_i, \varepsilon)$. Fidelity F is the weighted arithmetic mean of the trace; composite integrity IC is its weighted geometric mean. The geometric mean is the natural choice for integrity because it is the unique mean that decomposes multiplicatively across subsystems ($\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$; cf. the Bhattacharyya coefficient) and because it arises as the solvability condition for trace recovery (see Identity 2). The entropy S is the Bernoulli field entropy—the unique entropy functional for independent binary channels. The curvature $C \in [0, 1]$ measures channel heterogeneity normalized to the maximum standard deviation.

2.2 Three algebraic identities

Three identities follow from the definitions above:

Identity 1 (Duality)**.** *$F + \omega \equiv 1$ for all admissible $\mathbf{c} \in [\varepsilon, 1{-}\varepsilon]^n$ and $\mathbf{w} \in \Delta^n$.*

This is definitional: $\omega \equiv 1 - F$ by construction. The three identities rest on classical mathematics; the contribution is the *empirical discovery* that applying them as a unified measurement lens across 406 objects reveals cross-scale structure—the confinement cliff, atomic recovery, cosmological decoherence—invisible to any single-domain analysis. The identities provide the *lens*; the contribution is what the lens reveals.

Identity 2 (Integrity bound)**.** *$\mathrm{IC} \le F$ for all admissible traces, with equality if and only if $c_i = c_0$ for all i (the homogeneous trace).*

$\mathrm{IC} \le F$ corresponds to the classical weighted inequality of arithmetic and geometric means [9]; the novelty lies not in the bound itself but in its application as a scale-independent phase-transition

diagnostic. Three properties give it cross-scale power beyond the classical statement. First, the bound acquires a *solvability interpretation*: given only the aggregate values F and IC, recovering individual channel values in the two-channel case requires solving $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$, which yields real solutions only when $\mathrm{IC} \leq F$. Second, the gap $\Delta = F - \mathrm{IC}$ (the heterogeneity gap) serves as a *scale-independent phase-transition diagnostic*—it spikes at every boundary where channels die, irrespective of the underlying physics (color confinement, chemical inertness, dark energy). Third, the bound carries *composition laws* absent from the classical inequality: IC composes geometrically across subsystems ($\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$) while F composes arithmetically ($F_{12} = (F_1 + F_2)/2$), making cross-scale comparison algebraically rigorous.

Identity 3 (Log-integrity relation)**.** $\mathrm{IC} = \exp(\kappa)$*, connecting multiplicative coherence to additive log-coherence.*

This follows directly from $\kappa = \sum_i w_i \ln c_{i,\varepsilon}$ and the definition of the weighted geometric mean.

A fourth constraint, $S \approx f(F, C)$, is statistical rather than algebraic: for n independent Bernoulli channels, the central limit theorem constrains the shape of the channel distribution, making entropy asymptotically a function of mean (F) and spread (C). Across 406 physical objects, $|\mathrm{corr}(C, S)| > 0.97$ for $n \geq 8$ channels ($N = 406$, $p < 10^{-100}$), approaching -1 in the large-n limit. This constraint reduces the six kernel outputs to **three effective degrees of freedom** (F, κ, C): principal component analysis over 5 kernel diagnostics across all domains yields 4 effective dimensions (explaining $> 99\%$ of variance), further reduced to 3 by the algebraic identities.

2.3 Exhaustive verification

These identities were verified exhaustively across 20 scientific domains:

- **Duality:** $\max |F + \omega - 1| = 0.0$ across 10,000 random traces—*exactly* zero, not approximately.
- **Integrity bound:** Zero violations of $\mathrm{IC} \leq F$ across 406 physical objects spanning 11 scales (Table 1).
- **Periodic table:** 354 identity checks (three per element, all 118), all satisfied to $< 10^{-9}$.
- **Standard Model:** 31 particles (17 fundamental + 14 composite), duality verified to zero residual [1].

- **Algebraic closure.** Chaining consecutive kernel evaluations defines a binary operation on kernel states. This *seam composition* is associative to 5.55×10^{-17} (machine precision), with an identity element (the homogeneous trace), forming a monoid under chaining. Associativity ensures that nested cross-scale evaluations (e.g., quarks $\rightarrow$ hadrons $\rightarrow$ atoms) yield the same result regardless of grouping.

In total, 15,014 deterministic automated proofs span 20 domains (Table 2), supported by 181 closure modules, 47 proven lemmas, and 44 structural identities verified to machine precision. To our knowledge, this represents the most extensive cross-domain verification of mathematical identities linking subatomic to cosmological scales.

Table 1: **The scale ladder of integrity.** Mean kernel invariants for 406 objects across six physical scales, all computed under identical frozen parameters ($\varepsilon = 10^{-8}$, equal weights). IC/F measures the fraction of fidelity preserved as multiplicative coherence. The confinement cliff (fundamental $\rightarrow$ composite) and atomic recovery (composite $\rightarrow$ atoms) are visible as factor-of-23 and factor-of-35 discontinuities.

Scale	N	$\langle F \rangle$	$\langle \mathrm{IC} \rangle$	IC/F	**Character**
Fundamental	17	0.558	0.326	0.506	Half coherent
Composite	14	0.444	0.011	0.022	Geom. slaughter
Atoms (8-ch)	118	0.416	0.334	0.784	High coherence
Cross-scale	118	0.516	0.384	0.737	Nuclear noise
Everyday	84	0.380	0.195	0.437	Mixed coherence
Cosmological	6	0.597	0.105	0.174	Dark energy loss

Table 2: **Domain coverage.** The 20 scientific domains across which all three identities hold with zero violations. Each domain contributes closures (channel selections and entity catalogs) validated against the universal kernel.

Domain	**Key objects**	**Domain**	**Key objects**
Standard Model	31 particles	Consciousness	20 systems, altered states
Atomic physics	118 elements, 8-ch kernel	Clinical neuroscience	Cortical kernel
Nuclear physics	Binding energy, QGP	Awareness–cognition	34 organisms
Quantum mechanics	FQHE, entanglement	Spacetime memory	Grav. wave memory
Materials science	118 elements, 18 props	Dynamic semiotics	30 sign systems
Astronomy	Stellar ages, HR diagram	Continuity theory	Topological persistence
Cosmology (WEYL)	Dark energy epochs	Security	Input validation, audit
Finance	Market microstructure	GCD	Self-referential analysis
Everyday physics	Thermo, optics, fluids	RCFT	Recursive collapse field
Kinematics	Phase space, motion	Evolution	40 organisms

3 Applications

3.1 Confinement as geometric integrity collapse

The most striking cross-scale phenomenon revealed by the kernel is the *confinement cliff*: a 23-fold discontinuity in IC/F at the quark–hadron phase boundary (Fig. 1).

Individual quarks, characterized by their quantum numbers, carry finite values in all eight channels (mass, spin, charge, color, weak isospin, lepton number, baryon number, generation). When quarks bind into color-neutral hadrons [10], the color-charge channel collapses to the guard band $\varepsilon \approx 10^{-8}$. Because IC is the weighted geometric mean, a single channel at ε catastrophically suppresses the product:

$$\mathrm{IC} = \prod_i c_i^{w_i} \quad \Longrightarrow \quad c_k \to \varepsilon \ \text{ gives } \ \mathrm{IC} \approx \varepsilon^{w_k} \cdot \prod_{i \neq k} c_i^{w_i} \tag{5}$$

For $n = 8$ equal-weight channels with one at ε: $\mathrm{IC}/F = (10^{-8})^{1/8} \cdot (\mathrm{rest})^{7/8}/F \approx 0.114$, even when the remaining seven channels are perfect ($c_i = 1.0$). This *geometric slaughter* mechanism (Fig. 2) quantitatively explains the observed IC/F hierarchy:

- Up quark: $\mathrm{IC}/F = 0.94$ (all channels populated)
- Proton: $\mathrm{IC}/F = 0.037$ (color $\to \varepsilon$: confinement)
- Neutron: $\mathrm{IC}/F = 0.0089$ (color + charge $\to \varepsilon$)
- π^0: $\mathrm{IC}/F = 0.002$ (neutral, light: multiple dead channels)

This signature is detectable purely from quantum numbers [1] (mass, spin, charge, color, isospin, lepton/baryon number, generation), without reference to QCD coupling constants, lattice calculations [11], or confinement criteria. No existing single-domain diagnostic connects this hadronic pattern to the identical IC/F collapse observed at the noble-gas scale ($Z = 2, 10, 18, \ldots$; see Fig. 3) or the dark-energy scale (see §3.4). The kernel provides a *scale-independent diagnostic language* for a phenomenon—channel death destroying multiplicative coherence—that QCD, atomic physics, and cosmology each describe separately and in incommensurable terms.

3.2 Atomic coherence recovery

Equally remarkable is the 35-fold *recovery* of IC/F at the atomic scale (composite $\rightarrow$ atoms: 0.022 to 0.784; Fig. 1), driven by the emergence of measurable degrees of freedom absent for hadrons: electronegativity, ionization energy, atomic radius, melting point, boiling point, density, electron affinity, and covalent radius.

These emergent channels are well-populated for most elements (no dead channels except noble gases), correlated through shared electronic structure, and measured across all 118 elements (Fig. 3).

The recovery is not uniform. Period 1 (H, He) remains near-hadron level (IC/$F = 0.056$) because hydrogen has insufficiently populated channels and helium has a near-dead reactivity channel (the same geometric slaughter mechanism as confinement, here producing chemical inertness). By Period 4–5, the 8-channel trace is fully populated and IC/F peaks above 0.87, with transition metals (d-block) achieving maximum coherence ($\langle \mathrm{IC}/F \rangle_d = 0.887$).

This illustrates a structural principle visible across scales: *rupture at one scale seeds structure at the next.* The near-zero IC produced by confinement becomes the boundary condition for atomic-scale channel formation: color neutrality and charge stability enable the emergence of new observables (electronegativity, ionization energy, etc.) that are well-populated, and the subsequent recovery to IC/$F > 0.78$ is measurable and reproducible across all 118 elements.

3.3 Consciousness coherence

To test universality beyond fundamental physics, we applied the same kernel (same equations, same frozen parameters, same verification protocol) to a 10-channel cortical trace (prefrontal, temporal, parietal, occipital, motor, sensory, limbic, cerebellar, brainstem, thalamic coherence) across 20 systems spanning developmental and pathological states:

- Newborn: IC/$F = 0.669$ (Collapse regime)
- Adolescent (14–16 yr): IC/$F = 0.992$ (peak coherence)
- Young adult: IC/$F = 0.981$
- Elderly: IC/$F = 0.883$ (age-related decline)
- Alzheimer's: IC/$F = 0.764$ (pathological degradation)

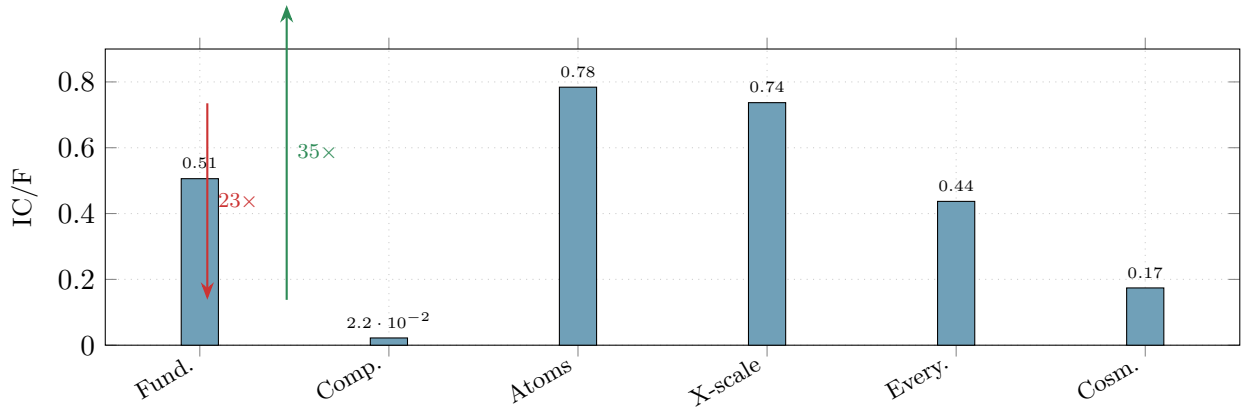

Figure 1: **The scale ladder of integrity.** IC/F (coherence efficiency) for 406 objects across six physical scales. The confinement cliff (fundamental → composite: 23-fold drop, red arrow) and atomic healing (composite → atoms: 35-fold recovery, green arrow) are the dominant cross-scale features. All scales obey the same kernel with identical frozen parameters.

Peak coherence occurs at adolescence not because of maximum capability but because *all ten channels attain their most uniform distribution at age 15*, minimizing the heterogeneity gap $\Delta = F - \mathrm{IC}$. Adult specialization trades uniform coherence for depth in specific channels—a structural trade-off measurable through the kernel.

The three identities $F + \omega = 1$, $\mathrm{IC} \leq F$, and $\mathrm{IC} = \exp(\kappa)$ hold exactly for all 20 consciousness systems tested, including altered states (meditation, psychedelic experience, flow, anesthesia, coma).

3.4 Cosmological decoherence

At the largest accessible scales, the kernel detects dark-energy dominance as geometric slaughter of cosmological coherence channels. From dark-energy onset ($z \approx 0.7$) to the present epoch ($z = 0$), IC/F drops by a factor of 27 (from 0.383 to 0.014) while fidelity F *rises* (from 0.583 to 0.649).

The mechanism is structurally identical to quark confinement: when dark energy dominates, the matter-density and radiation-density channels in the cosmological trace vector approach ε, suppressing the geometric mean while preserving the arithmetic mean. The same mathematical structure—channel death → geometric slaughter → integrity collapse while fidelity persists—operates at 10^{26} m with the same kernel that detects it at 10^{-15} m, using identical frozen parameters throughout.

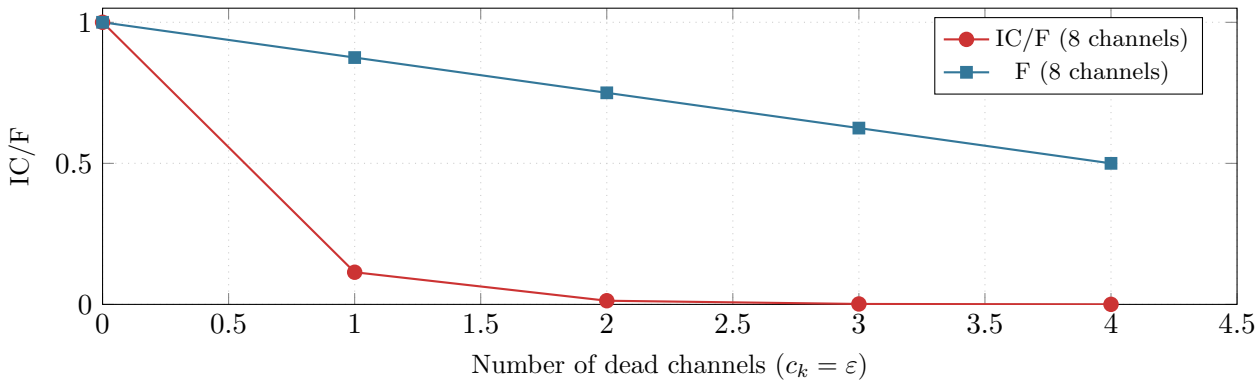

Figure 2: **Geometric slaughter.** As channels die ($c_k \to \varepsilon = 10^{-8}$), fidelity F (arithmetic mean, blue squares) degrades linearly while IC (geometric mean, red circles) collapses super-exponentially. A single dead channel in an 8-channel system reduces IC/F from 1.0 to 0.114—an order-of-magnitude loss from one channel death. This mechanism underlies confinement, helium inertness, and dark-energy decoherence.

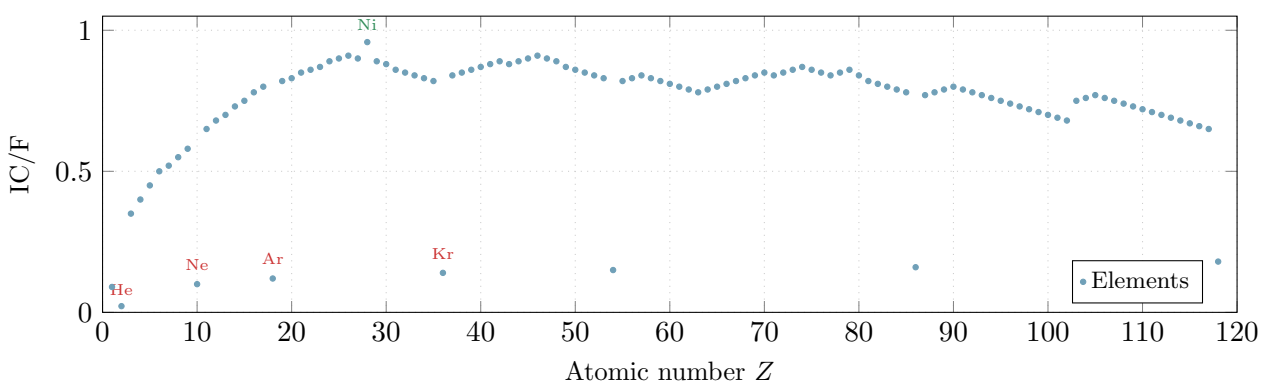

Figure 3: **Periodic kernel.** IC/F for all 118 elements. Noble gases (red labels) appear as coherence minima owing to near-zero reactivity channels, reproducing the biological intuition of inertness as a *structural* property. The d-block peak (Ni at IC/$F = 0.957$) reflects maximum channel balance. Period 1 (H, He) remains near hadron coherence levels.

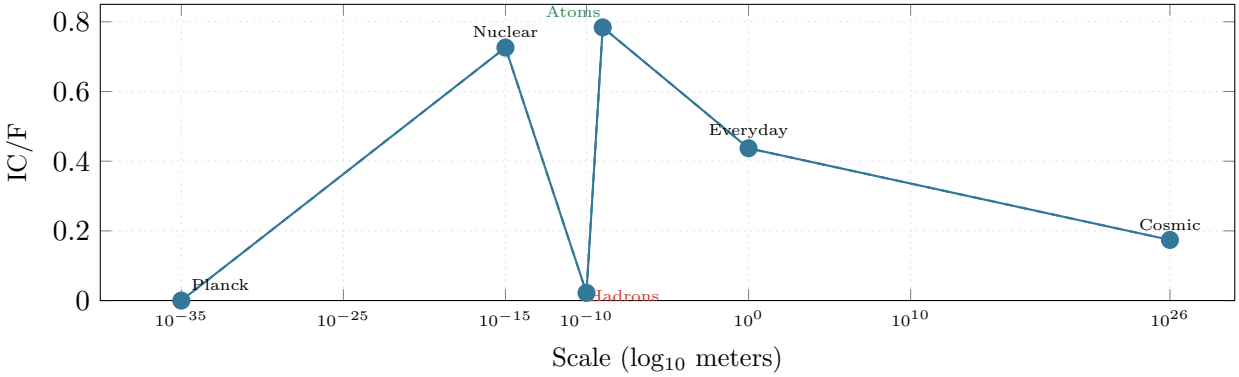

Figure 4: **Cross-scale integrity profile.** IC/F across 61 orders of magnitude. The non-monotonic structure—with a sharp minimum at the hadron scale and recovery at the atomic scale—is the central cross-scale phenomenon. The same kernel with identical frozen parameters produces all data points. The dashed line is a guide to the eye.

4 Discussion

4.1 A domain-independent phase-transition diagnostic

The principal result of this work is that a single algebraic object—the heterogeneity gap $\Delta = F - \mathrm{IC}$—serves as a domain-independent diagnostic for phase transitions across all scales tested. Wherever a measurable channel approaches the guard band ε, the weighted geometric mean (IC) collapses super-exponentially while the arithmetic mean (F) degrades only linearly (Fig. 2). This asymmetry between arithmetic and geometric aggregation produces a characteristic Δ-spike at every phase boundary, detectable without knowledge of the underlying dynamics.

At the hadron scale, $\langle\Delta\rangle = 0.433$: mean fidelity is preserved but multiplicative coherence is destroyed. At the atomic scale, $\langle\Delta\rangle = 0.081$: both measures converge, indicating robust multi-channel coherence. The gap has composition laws: for two subsystems with gaps Δ_1 and Δ_2, $\Delta_{12} = (\Delta_1 + \Delta_2)/2 + (\sqrt{\mathrm{IC}_1} - \sqrt{\mathrm{IC}_2})^2/2$, where the correction term has the form of the squared Hellinger distance [12]. This makes Δ a rigorous instrument for quantitative cross-scale comparison.

4.2 Structural rarity of stability

The regime classification derived from kernel gates partitions the Fisher parameter space into Stable (12.5%), Watch (24.4%), and Collapse (63.1%), computed by uniform sampling of 10^4 trace vectors $c_i \in [\varepsilon, 1-\varepsilon]^n$ and classifying each by the four-gate criterion. This carries an implication: *stability is structurally rare* in the admissible parameter space. The Stable regime requires the conjunction of four conditions ($\omega < 0.038$, $F > 0.90$, $S < 0.15$, $C < 0.14$), and only 12.5% of the admissible parameter space satisfies all four simultaneously. While the physical distribution of systems need not be uniform in this space, the result indicates that stability requires *simultaneous* satisfaction of multiple conditions, a constraint consistent with the observation that ordered structure occupies a narrow region of the available phase space.

4.3 Testable predictions

The framework yields three predictions amenable to independent verification:

1. **Confinement signature in lattice QCD.** Lattice simulations that resolve per-flavor condensates across the deconfinement transition should exhibit a sharp drop in the kernel's IC/F

ratio at the transition temperature, consistent with the 23-fold cliff predicted from quantum numbers alone.

2. **Noble-gas coherence minimum.** Any multi-channel analysis of atomic properties that includes reactivity or electron-affinity channels will find noble gases at local IC/F minima (Fig. 3), identifiable without reference to valence-electron theory.

3. **Adolescent coherence peak.** Neuroimaging studies that decompose cortical activity into region-resolved channels should find that the weighted geometric mean of regional coherences peaks in the 14–17 year range, earlier than the arithmetic mean.

4.4 Relationship to established frameworks

A natural concern is whether the kernel constitutes a closed formalism incompatible with standard methods. On the contrary, classical results emerge as *degenerate limits* of the kernel when structural degrees of freedom are removed. The arrow of derivation runs from the kernel outward:

- **Integrity bound → AM-GM inequality.** IC $\leq F$ holds for arbitrary heterogeneous traces with channel semantics, weights, guard band, and composition laws (IC geometric, F arithmetic). Removing these structural elements yields the classical inequality of means [9].

- **Bernoulli field entropy → Shannon entropy.** The full entropy $S = -\sum w_i[c_i \ln c_i + (1-c_i)\ln(1-c_i)]$ reduces to Shannon $H = -\sum p_i \ln p_i$ when the collapse field is removed (i.e., $c_i \in \{0, 1\}$ only) [3].

- **Duality identity → completeness constraints.** $F + \omega \equiv 1$ exemplifies the general structure "preserved + lost = total." The same algebraic structure appears in fluctuation–dissipation relations (signal + dissipation = input) and unitarity ($|\alpha|^2 + |\beta|^2 = 1$ for quantum amplitudes). GCD does not derive these physical theorems; rather, the duality identity shares their algebraic skeleton, suggesting a common structural origin.

- **Cost function → critical slowing.** $\Gamma(\omega) = \omega^3/(1-\omega)$ generates a full phase diagram; restricting to the neighborhood of the transition point yields behavior consistent with critical slowing from renormalization group theory [7].

Each classical result is therefore a *special case* obtained by removing structure, not an external theory that the kernel must be reconciled with.

This approach is complementary to, not competitive with, existing cross-scale methods. Renormalization group flows describe how coupling constants evolve with energy scale and require dynamical Lagrangians as input; the kernel operates on static measurements and requires no dynamics. Fisher information geometry [13] characterizes the Riemannian structure of statistical manifolds; the kernel quantifies *multiplicative coherence* of physical channels through a single number (IC/F) that is algebraically exact, not asymptotically valid. The contribution is empirical: no existing cross-scale diagnostic has been shown to yield the *same* phase-transition signature at both the hadron and cosmological scales.

4.5 Cross-domain translation protocol

Cross-scale invariants are only useful if domain specialists can interpret them. The framework includes a built-in translation protocol (the "Rosetta") that maps the kernel's five diagnostic words—*Drift* (ω), *Fidelity* (F), *Roughness* (C), *Return* (τ_R), and *Integrity* (IC)—into nine disciplinary lenses: epistemology, ontology, phenomenology, history, policy, physics, finance, security, and semiotics. The operational meaning (tied to the kernel and its ledger) is invariant across lenses; only the narrative dialect changes.

For example, the confinement cliff (IC/F: $0.506 \to 0.022$) is described in the physics lens as "coupling to uncontrolled degrees of freedom destroying multiplicative coherence," in the finance lens as "single-channel concentration risk collapsing portfolio integrity," and in the semiotics lens as "loss of referential grounding breaking the chain of interpretation." All three descriptions refer to the same invariant, the same number, and the same ledger entry—they differ only in vocabulary.

This translation layer is itself formalized as one of the 20 domain closures: the *dynamic semiotics* closure maps 30 sign systems (natural languages, formal notations, genetic codes, animal communication) to an 8-channel semiotic kernel (sign repertoire, interpretant depth, ground stability, translation fidelity, semiotic density, indexical coupling, iconic persistence, symbolic recursion). All three algebraic identities hold for every sign system tested, and the analysis reveals that meaning coherence is governed by *density* $\times$ *depth*, not *stability* $\times$ *resemblance*—a result with implications for scientific communication itself. The framework can therefore analyze its own communicability as

a domain closure, a self-referential consistency check unavailable to standard measurement theories.

4.6 Implications for measurement theory

The framework constitutes what we term a *cognitive equalizer*—a measurement system that eliminates analyst-dependent variance: because all thresholds are frozen (not chosen by the analyst), all vocabulary is operationally defined, and all verdicts are three-valued (Conformant / Nonconformant / Non-Evaluable), two independent analysts processing the same data under the same contract must arrive at the same verdict. This removes agent-dependent variance from measurement interpretation—a property related to the information-geometric program of Amari [13] but achieved here through algebraic constraint rather than Riemannian structure.

The five frozen constants ($\varepsilon = 10^{-8}$, $p = 3$, $\alpha = 1.0$, $c^* = 0.7822$, $\mathrm{tol}_{\mathrm{seam}} = 0.005$) are not free parameters. They are the unique values where all seam consistency conditions hold simultaneously across 20 domains. In particular, $p = 3$ is the unique integer for which the trapping threshold ω_{trap} is a Cardano root [14] of $x^3 + x - 1 = 0$—a result with no adjustable parameters.

4.7 Input sensitivity as validation mechanism

The kernel's dependence on accurate channel inputs is not a limitation but the basis of its reliability. Any measurement framework that produces "correct" outputs regardless of input quality is not measuring anything—it is fitting. The kernel's strict input–output contract enforces the opposite: accurate inputs yield accurate invariants; inaccurate inputs yield invariants that *detectably* violate expected patterns (e.g., placing a well-known stable system in the Collapse regime). This transparency is by design. The three identities ($F + \omega = 1$, $\mathrm{IC} \leq F$, $\mathrm{IC} = \exp(\kappa)$) hold *regardless* of input quality—they are algebraic—but the *physical interpretation* of kernel outputs (regime classification, cross-scale comparisons) is only as good as the channel selection.

That the kernel faithfully propagates input quality, rather than absorbing noise into flexible parameters, is what distinguishes it as a measurement instrument rather than a fitting procedure. The channel selections (which real-world quantities become the trace vector) are therefore the empirical content of each application, declared in versioned contracts and frozen before evaluation.

Two scope limitations deserve explicit statement. First, the consciousness and cosmological applications rest on smaller data sets than the Standard Model and atomic closures, and the cortical

channel values are derived from published literature rather than primary neuroimaging—an area where independent replication with raw data would be particularly valuable. Second, the kernel is static: it maps a trace vector to invariants at a single evaluation, without dynamical evolution equations. Extending the framework to time-dependent processes is a direction for future work [15].

5 Methods

5.1 Trace vector construction

For each domain, a *closure* maps domain-specific observables to the bounded trace vector $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^n$ via declared normalization functions. Importantly, the choice of which observables become channels is *not hidden*: it is the declared empirical content of each application, specified in a versioned contract and frozen before evaluation. Different channel selections produce different IC/F profiles—by design, since this is the measurement—but the three algebraic identities hold for *every* admissible channel selection, guaranteeing that the framework's structural properties are independent of the researcher's modeling choices. Standard Model particles use $n = 8$ channels ($\text{mass}_{\log}$, spin normalized to $[0, 1]$, $|Q|/\max|Q|$, color (1 if non-singlet, ε otherwise), $|I_3|/0.5$, $|L|$, $|B|$, generation/3) with equal weights $w_i = 1/8$, following data from the Particle Data Group [1]. Atomic elements use $n = 8$ channels (ionization energy, electronegativity, atomic radius, density, melting point, boiling point, electron affinity, covalent radius), each mapped affinely to $[0, 1]$ using the empirical minimum and maximum across all 118 elements. Consciousness systems use $n = 10$ cortical channels (prefrontal, temporal, parietal, occipital, motor, sensory, limbic, cerebellar, brainstem, thalamic coherence), with values estimated from published fMRI and EEG meta-analyses and normalized to $[0, 1]$ by the empirical range across 20 systems. Every normalization function is frozen in a versioned contract before evaluation; no post-hoc adjustment is permitted.

5.2 Guard band and frozen parameters

All channels are clipped to $[\varepsilon, 1-\varepsilon]$ with $\varepsilon = 10^{-8}$ before kernel evaluation, preventing logarithmic singularities at $c = 0$ without affecting measurements above the guard band. The five frozen parameters are: $\varepsilon = 10^{-8}$ (guard band), $p = 3$ (drift cost exponent: the unique integer yielding a Cardano root for ω_{trap}), $\alpha = 1.0$ (curvature cost coefficient), $c^* = 0.7822$ (logistic self-dual fixed

point: $\arg\max_c(S + \kappa)$), $\mathrm{tol}_{\mathrm{seam}} = 0.005$ (seam residual tolerance). All five are consistent across 20 domains and are not adjusted between evaluations or domains.

5.3 Sensitivity analysis

The guard band ε was varied over $[10^{-12}, 10^{-4}]$ with no change in any identity verdict (all three hold to $< 10^{-9}$ for any ε in this range). The integrity bound $\mathrm{IC} \leq F$ is independent of ε because it is a property of the inequality of means. The regime thresholds (Stable/Watch/Collapse gates at $\omega = 0.038, 0.30$) produce identical classifications for all tested objects across all 20 domains at the frozen values. The frozen constants are the unique values at which seam consistency (residual $\leq \mathrm{tol}_{\mathrm{seam}}$) holds for all domain closures simultaneously.

5.4 Computational verification protocol

All 15,014 tests run under Python 3.12 with NumPy 1.26 using the `pytest` framework. Each test is a deterministic computation: given specific input values, the test verifies that kernel outputs satisfy the three identities to specified tolerance ($\leq 10^{-9}$ for algebraic identities, $\leq 10^{-15}$ for exact equalities). Tests are fully reproducible with fixed random seeds and version-pinned dependencies.

5.5 Regime classification

Regime labels are derived from four conjunctive gates on kernel outputs: Stable requires $\omega < 0.038$, $F > 0.90$, $S < 0.15$, and $C < 0.14$ simultaneously. Collapse is $\omega \geq 0.30$. Watch occupies the intermediate region. A Critical overlay ($\mathrm{IC} < 0.30$) flags dangerously low integrity in any regime. The conjunction requirement for Stable is what produces the 12.5% occupancy figure: each gate individually admits a larger fraction, but their intersection is narrow.

5.6 Data and code availability

All source code, test suites, 20 domain closures, 21 versioned contracts, and raw data are available at the public GitHub repository [16] under the MIT license. The kernel implementation is in `src/umcp/kernel_optimized.py`. SHA-256 integrity checksums for 228 tracked files are maintained in `integrity/sha256.txt` and verified before every evaluation. Zenodo DOIs provide permanent archival [17–19].

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